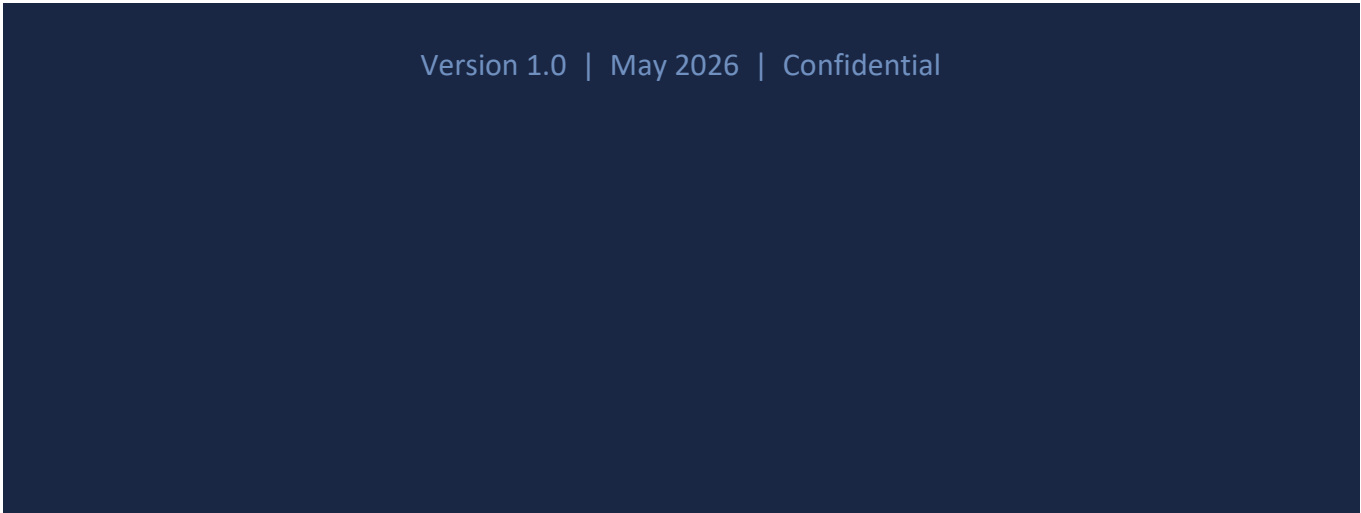




Property Portal · Admin Panel · Agent CRM



Version 1.0 | May 2026 | Confidential

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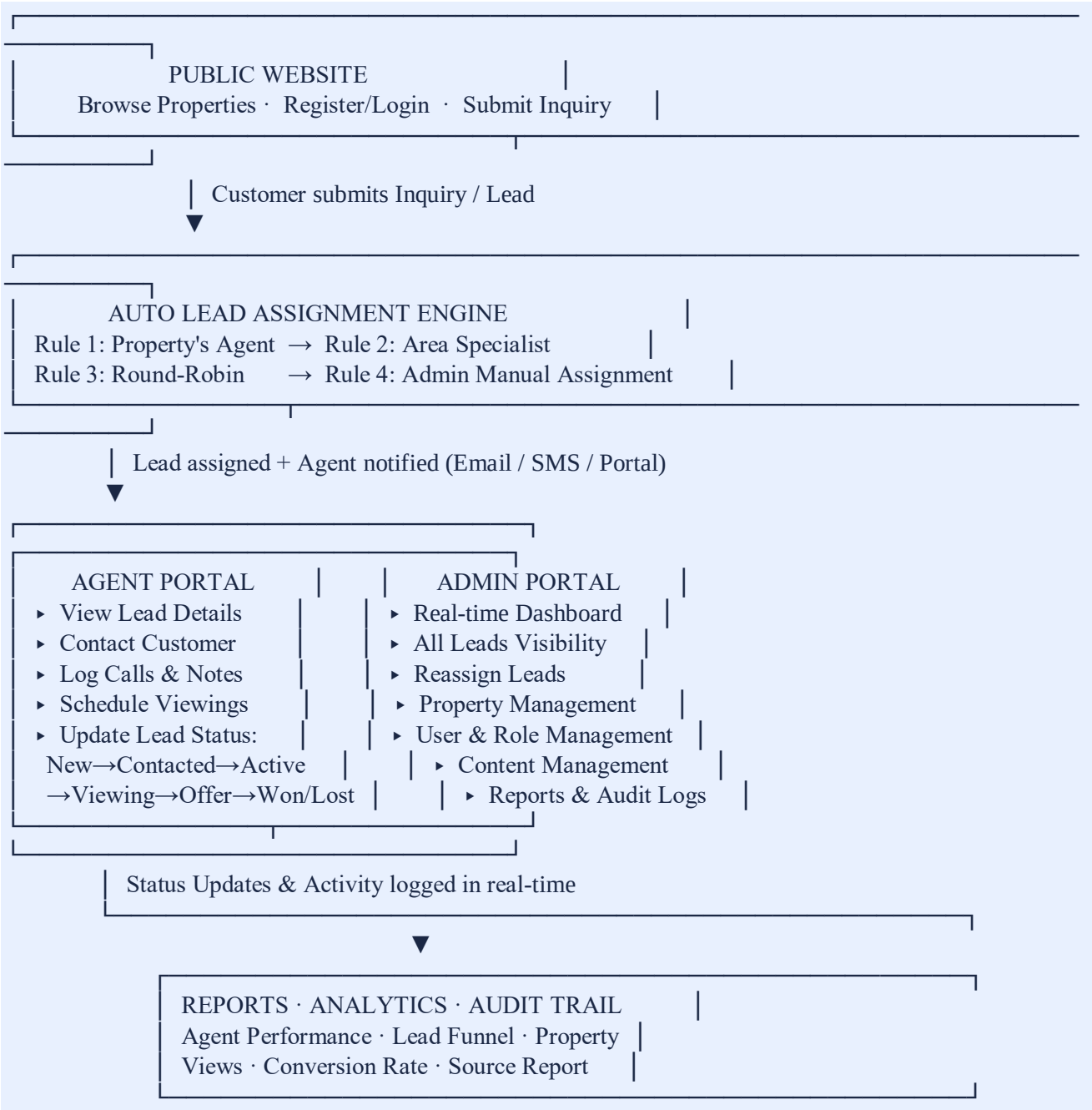
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SYSTEM FLOWCHART

The diagram below illustrates the complete end-to-end data and user flow across the Public Website, Auto Lead Assignment Engine, Agent Portal, and Admin Portal.



Flowchart Legend

■ Public Website ■ Agent Portal ■ Admin Portal ■ System Events (Lead Created, Assigned, Updated)

PAGE 1 — SYSTEM OVERVIEW, ARCHITECTURE & USER ROLES

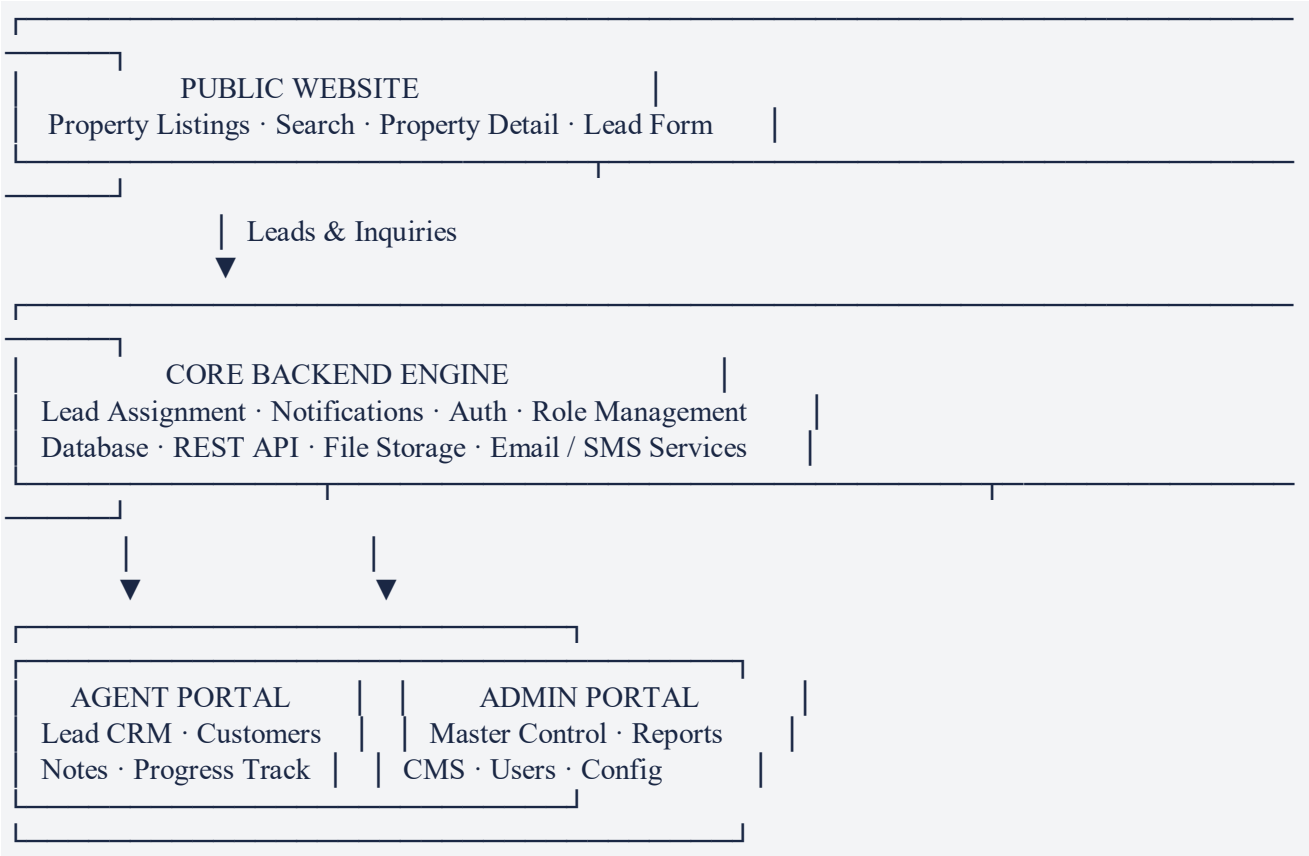
1.1 Project Overview

This document defines the complete specification for a Real Estate Property Portal — a full-stack platform comprising a public-facing property website, a role-based admin panel, and a dedicated agent portal. The platform is designed to streamline property discovery, lead generation, lead management, and inter-team coordination for a real estate business.

The system enables Customers to browse, search, and inquire about properties online; Agents to receive, manage, and close leads through a dedicated CRM-style portal; and Admins to oversee all platform activity, manage content, configure the system, and generate comprehensive reports.

1.2 System Architecture Overview

The platform consists of three interconnected layers communicating through a secure backend API:



Recommended Technology Stack

Layer	Technology
Frontend (Website)	Next.js (React) + Tailwind CSS
Admin & Agent Portal	React.js + Ant Design / Material UI
Backend API	Node.js (Express) or Laravel (PHP)
Database	PostgreSQL (relational) + Redis (cache/sessions)
File / Image Storage	AWS S3 or Cloudflare R2
Authentication	JWT + Role-Based Access Control (RBAC)

Layer	Technology
Email Notifications	SendGrid / Postmark
SMS Notifications	Twilio
Deployment	Docker + AWS / DigitalOcean

1.3 User Roles & Permissions Matrix

The platform uses Role-Based Access Control (RBAC) with four primary roles. Each role has scoped access to platform features, configurable by the Super Admin through the Permissions Manager.

Role Definitions

- Super Admin — Full platform control, all permissions. Access: Admin Portal
- Admin — Operational admin, limited system configuration. Access: Admin Portal
- Agent — Sales representative managing leads & listings. Access: Agent Portal
- Customer — End-user browsing & submitting inquiries. Access: Public Website

Permissions Matrix

Feature / Module	Super Admin	Admin	Agent	Customer
View Public Website	✓	✓	✓	✓
Submit Property Inquiry	✓	✓	✓	✓
Add / Edit Properties	✓	✓	✓(assigned)	✗
Delete Properties	✓	✓	✗	✗
Publish / Unpublish	✓	✓	✗	✗
View All Leads	✓	✓	✗	✗
View Assigned Leads	✓	✓	✓	✗
Update Lead Status	✓	✓	✓	✗
Reassign Leads	✓	✓	✗	✗
Create / Edit Users	✓	✓	✗	✗
Delete Users	✓	✗	✗	✗
Assign Roles	✓	✓	✗	✗
Edit Website Pages (CMS)	✓	✓	✗	✗
View All Reports	✓	✓	✗	✗
View Own Performance	✓	✓	✓	✗
View Audit Logs	✓	✗	✗	✗
System Configuration	✓	✗	✗	✗
Role & Permission Config	✓	✗	✗	✗

PAGE 2 — PUBLIC WEBSITE MODULES & FEATURES

2.1 Website Structure & Navigation

The public website is the customer-facing layer of the platform. It is fully SEO-optimized, mobile-responsive, and dynamically driven by content managed through the Admin Portal CMS.

[Home](#) | [Properties](#) | [Off-Plan](#) | [Agents](#) | [About Us](#) | [Blog](#) | [Contact](#)

2.2 Home Page

The homepage serves as the primary entry point and conversion funnel for all visitors.

- Hero Banner — Full-width image/video slider with headline and CTA buttons (Buy, Rent, Sell)
- Smart Search Bar — Prominent search with filters: Type, Location, Price Range, Bedrooms, Status
- Featured Properties — Curated property cards selected and managed by admin
- Property Categories — Apartments, Villas, Townhouses, Offices, Retail, Plots
- Featured Projects — Off-plan / new launch developments with special styling
- Stats Section — Dynamic counters: Total Listings, Years of Experience, Happy Clients, Agents
- Featured Agents — Agent cards with photo, name, specialization, and contact button
- Testimonials / Reviews — Customer reviews carousel managed via CMS
- Latest Blog / News — Recent articles automatically pulled from the blog module
- Footer — Company info, quick links, social media, newsletter signup, contact details

2.3 Property Listings Page

The main property search and browsing interface with advanced filtering and multiple view modes.

Search & Filter Panel:

- Keyword search (area, building name, project name, reference number)
- Property Type: Apartment, Villa, Townhouse, Office, Retail, Plot
- Status: For Sale / For Rent / Off-Plan
- Price range slider (min–max with dynamic currency formatting)
- Bedrooms / Bathrooms count selector
- Area range (sq ft / sq m)
- Furnishing status: Furnished / Unfurnished / Partly Furnished
- Community, Sub-community, Building location taxonomy
- Amenities multi-select (Pool, Gym, Parking, Pet-friendly, etc.)
- Completion status: Ready / Off-Plan

Display & Interaction:

- View Toggle — Grid view / List view / Map view
- Sort Options — Price (Low–High / High–Low), Newest First, Most Viewed
- Pagination with infinite scroll option
- Save to Wishlist — Logged-in customers can bookmark properties
- Property Card — Thumbnail, title, price, beds/baths, size, location, agent avatar, listing badge
- Map Integration — Interactive Google Maps / Mapbox with property pins and cluster markers

2.4 Property Detail Page

The individual listing page is the primary conversion point for leads. It provides all property information and a direct inquiry channel.

- Image Gallery — Full-screen carousel with photo, video, and virtual tour support
- Property Title & Key Stats — Price, beds, baths, area, type, reference number, status
- Tabbed Sections — Overview, Details, Amenities, Location, Payment Plan, Floor Plan
- Location Tab — Embedded map with nearby landmarks (schools, malls, metro stations)
- Agent Card — Assigned agent photo, name, phone, WhatsApp direct link, email contact
- Inquiry Form — Pre-filled with property reference; submission creates a lead record
- Similar Properties — Related listings carousel at bottom of page
- Social Share & Wishlist Save — Share to WhatsApp, email, social media; save to account

Property Form Fields (Admin / Agent):

Section	Fields
Basic Info	Title, Reference No., Property Type, Listing Type (Sale/Rent), Status
Description	Short description, Full rich-text description
Location	Country, City, Community, Sub-community, Building, Address, Map pin
Details	Price, Bedrooms, Bathrooms, Area (sqft/sqm), Floor, View, Parking, Furnishing
Amenities	Multi-select checklist (Pool, Gym, Security, Kids Area, etc.)
Media	Photo gallery (drag-drop reorder), Video URL, Virtual tour URL
Floor Plan	Upload floor plan image or PDF
Payment Plan	Milestone table — % + amount + date (for off-plan properties)
SEO	Meta title, meta description, custom URL slug
Visibility	Featured toggle, Homepage feature, Listing badge (New/Reduced/Hot)
Permit / RERA	Permit number, developer name, completion date

2.5 Customer Registration & Authentication

Registration:

- Sign up with: Full Name, Email Address, Phone Number, Password
- Social login options: Google / Facebook / Apple ID (configurable)
- Email verification link sent on registration
- Optional phone OTP verification for added security

Customer Dashboard (My Account):

- Profile management: name, phone, profile photo, property preferences
- Saved / Wishlist properties with quick-view and remove options
- My Inquiries — list of all submitted leads with current status and agent assigned
- Notification preferences: email, SMS, WhatsApp toggle per event type

2.6 Agents Directory

- Searchable, filterable list of all published agents
- Agent card: Photo, Name, Specialization, Languages, Listing count, Rating
- Individual Agent Profile Page with bio, active listings, and direct contact form
- Contact form on agent page creates a lead tagged directly to that agent

2.7 Blog / News Module

- Article listing with category filter and keyword search
- Rich article pages with images, author attribution, date, social sharing
- Related articles section, comment section (optional / moderated)
- SEO metadata per article: meta title, description, canonical URL, OG image

2.8 Static Pages & Contact

- CMS-managed pages: About Us, Why Choose Us, Careers, Privacy Policy, FAQs, Terms
- Contact Page: office address, phone, embedded map, general inquiry form
- General inquiry form creates a lead of type 'General Inquiry'
- Site-wide WhatsApp floating button (click-to-chat with configured number)

PAGE 3 — ADMIN PORTAL: FULL FEATURE SET

The Admin Portal is the master control center of the platform. It provides full visibility across all operations with tools for managing properties, users, leads, content, and system configuration. Access is restricted to Admin and Super Admin roles with 2FA support.

3.1 Admin Dashboard

The main landing screen — a real-time operations overview of the entire platform.

KPI Cards:

- Total Properties: Published / Draft / Pending Review breakdown
- Total Leads: Today / This Week / This Month with trend indicators
- New Unassigned Leads: Requires immediate attention (highlighted in amber)
- Active Agents: Currently online vs. total count
- Registered Customers: Total with monthly growth
- Conversion Rate: Lead → Viewing → Closed funnel percentage

Dashboard Widgets:

- Lead Pipeline Chart — Visual funnel: New → Active → Viewing → Negotiation → Closed/Lost
- Recent Leads Table — Last 10 leads with status, source, and assigned agent
- Agent Performance Snapshot — Top 5 agents by leads closed this month
- Property Views Chart — Bar chart of most viewed properties this week
- Activity Feed — Real-time log: new lead, property updated, user created
- Quick Actions Panel — Add Property, Create User, View Reports shortcuts

3.2 Property Management

Full lifecycle management of all property listings from draft to archive.

Property Status Workflow:

Draft → Pending Review → Published → [Archived / Sold / Rented]

Listing View & Bulk Actions:

- Table: Thumbnail, Title, Type, Location, Price, Status, Agent, Date, Actions
- Search by title or reference number; filter by type, status, agent, area, price
- Bulk actions: Publish, Unpublish, Archive, Delete, Reassign Agent

3.3 Lead Management

Central hub for all leads generated from the website with full assignment and tracking controls.

Lead Status Pipeline:

New → Contacted → Active → Viewing Scheduled → Viewing Done
→ Offer Made → Negotiation → Won (Closed) ✓ / Lost ✗

Auto-Assignment Rules (configurable by Admin):

- Rule 1: Property already has an assigned agent → assign to that agent
- Rule 2: Lead area matches an agent's area specialization → lowest-load matching agent
- Rule 3: Round-robin rotation among available agents respecting lead cap per agent

- Rule 4: Fallback → assign to Admin / Team Leader for manual assignment

Lead Detail View Includes:

- Customer information and contact details
- Property of interest (linked to active listing)
- Lead source and UTM/referral tracking
- Assigned agent with reassign button
- Full activity timeline with timestamps and actor names
- Admin notes (visible to admin only)
- Manual status override capability

3.4 User & Role Management

Users Table:

- All users across all roles: Name, Email, Phone, Role, Status, Date Joined, Last Login
- Filter by Role, Status, Date; search by name, email, or phone

Create / Edit User:

- Name, Email, Phone, Password (auto-generate or manual set)
- Role assignment: Super Admin / Admin / Agent / Customer
- For Agents: Designation, Languages, Bio, Specialization, WhatsApp, Areas
- Account status: Active / Inactive / Suspended with audit trail
- Send welcome email with login credentials on creation

Role & Permission Management:

- Visual permission matrix editor — toggle individual permissions per role
- Create custom roles (e.g., 'Senior Agent', 'Marketing Admin', 'Branch Manager')
- Save and immediately apply changes across all users of that role

3.5 Content Management System (CMS)

Homepage Management:

- Hero banner management: add/edit/delete banners, set order, configure CTA links
- Featured Properties selection: pin specific listings to homepage sections
- Stats counters: manually set or auto-calculate from database
- Testimonials: add, edit, delete, sort, publish/unpublish individual reviews

Pages Management:

- Edit static pages using WYSIWYG rich text editor (About Us, FAQs, Privacy Policy)
- Per-page SEO: meta title, description, OG image, canonical URL

Blog / News Management:

- Create, edit, publish, unpublish, delete articles with rich-text editor
- Categories and tags management; author assignment; featured image upload

Navigation & Taxonomy:

- Edit main menu links and order; manage footer links and social media links
- Manage amenities (icons + labels), property types, communities, sub-communities

Announcements & Pop-ups:

- Site-wide announcement bar (top of website) with configurable display rules
- Pop-up modals: on-exit intent, timed, or page-scroll triggers for lead capture

3.6 Reports & Analytics

Report	Description
Lead Summary Report	Total leads by date range, source, status, and property type
Agent Performance Report	Leads assigned, contacted, closed, conversion rate per agent
Property Performance Report	Views, inquiries, and conversion rate per listing
Conversion Funnel Report	Drop-off analysis at each stage of the lead pipeline
Customer Activity Report	Registrations, logins, saved properties, and inquiries
Revenue / Deals Report	Closed deals, deal value, commission tracking (if applicable)
Source Attribution Report	Leads by source: organic, WhatsApp, direct, social media, portal

Report Features:

- Date range filter: custom range or presets (Today, Week, Month, Quarter, Year)
- Export to PDF and Excel / CSV formats
- Scheduled email delivery: daily, weekly, monthly to selected admin emails
- Interactive charts: Bar, Line, Pie, Funnel (powered by chart library)

3.7 Audit Logs

Full trail of all actions taken within the platform — accessible to Super Admin only.

Logged Event	Source
User login / logout / failed attempts	All portals
Property created / edited / deleted / published	Admin & Agent portals
Lead created / assigned / reassigned / status changed	All portals
User created / edited / deleted / role changed	Admin portal
Content changes (CMS edits)	Admin portal
System configuration changes	Admin portal (Super Admin only)
Report exports	Admin portal

Log Table columns: Timestamp, User (actor), Action Type, Module, Details (before → after), IP Address. Filter by user, action type, date range, or module. Export to CSV for compliance review.

3.8 System Configuration

General Settings:

- Website name, logo, favicon, contact email, phone, WhatsApp, office address
- Default currency, measurement unit (sqft / sqm), time zone

Email & Notification Settings:

- SMTP configuration for outbound email delivery
- Email templates: lead assigned, status changed, welcome, inquiry confirmation

- SMS gateway configuration (Twilio API keys) and WhatsApp Business API integration

SEO & Analytics Settings:

- Google Analytics ID, Google Tag Manager ID, Facebook Pixel ID
- Sitemap auto-generation toggle, Robots.txt editor

Integration Settings:

- Property portal API sync: Bayut, Property Finder auto-syndication
- CRM integration: Salesforce, HubSpot (optional)
- Payment gateway for premium/featured listing fees

Backup & Maintenance:

- Automated database backup schedule (daily, retained 30 days)
- Maintenance mode toggle — shows custom coming-soon page to public visitors

PAGE 4 — AGENT PORTAL: FEATURES & WORKFLOW

The Agent Portal is a dedicated, focused workspace for real estate agents. Designed as a lightweight CRM, it enables agents to manage their assigned leads, track customer interactions, and update deal progress. Agents only ever see their own data — cross-agent visibility is restricted by the RBAC layer.

4.1 Agent Dashboard

Dashboard Widgets:

- My Lead KPIs: Total assigned leads, New (uncontacted), Active, Viewings this week, Closed this month
- Lead Pipeline View — Kanban board or funnel chart showing own leads by status stage
- Today's Tasks / Follow-ups — Scheduled reminders sorted by urgency
- Upcoming Viewings — Calendar widget showing all scheduled property viewings
- My Performance Stats — Leads contacted rate, conversion rate, avg. time-to-first-contact
- Recent Activity — Last 10 actions taken on any of agent's leads
- Notifications Bell — New lead assigned, status reminders, admin messages

4.2 Lead Management (Agent View)

The core feature of the Agent Portal — a personal CRM for all leads assigned to the agent.

Leads Table:

- Columns: Customer Name, Phone, Email, Property Interested In, Status, Date Assigned, Priority
- Filter by Status, Date, Property Type, Priority flag; search by name, phone, or email

Lead Detail Page Sections:

Customer Info:

- Name, phone, email, preferred contact method; customer requirements and budget notes

Lead Status Pipeline:

New → Contacted → Active → Viewing Scheduled → Viewing Done
→ Offer Made → Negotiation → Won (Closed) ✓ / Lost ✗

- One-click status update buttons with confirmation prompt
- Loss reason dropdown required when marking as Lost (for reporting)
- Expected closing date capture for Won / Negotiation status leads

Activity Timeline:

- Chronological log of all actions: status changes, calls, notes, emails, viewings
- Timestamps and actor name on every timeline entry; immutable once saved

Available Actions:

- Add Note — Internal note visible to agent and admin only
- Log Call — Record date/time, duration, outcome, next follow-up date
- Send Email — Compose and send email directly from portal; auto-logged in timeline
- Schedule Viewing — Date/time, property, customer confirmation status
- Set Follow-up Reminder — Date + reminder note; appears in Today's Tasks dashboard widget
- Upload Documents — Offer letters, signed agreements, ID copies; stored securely

Priority Flags:

- Mark leads as Hot / Warm / Cold for personal pipeline prioritization

4.3 Customer Management (Agent View)

- List of all unique customers linked to the agent's leads
- Customer profile: full contact info, all linked leads, complete interaction history
- Edit customer notes: budget, property preferences, timeline, family size requirements
- Update contact preferences: best time to call, preferred channel (call/email/WhatsApp)

4.4 My Properties

- View all property listings assigned to the agent with inquiry/lead count per listing
- Edit property details within admin-configured allowed fields
- Mark property as 'Under Offer' or 'Sold/Rented' — triggers admin review notification
- Upload new photos or update property details (subject to admin publish approval)

4.5 Calendar, Performance & Profile**Agent Calendar:**

- Monthly / weekly calendar view of all viewing appointments and follow-up reminders
- Auto-populated from 'Schedule Viewing' and 'Set Reminder' actions in lead detail
- Manual event addition; optional Google Calendar sync integration

My Performance:

- Leads assigned this month vs. last month (trend arrow)
- Leads contacted within 1 hour / 24 hours / 48 hours of assignment
- Conversion rate: Lead → Won; viewings conducted; deals closed with value
- Monthly trend charts for personal goal tracking

Agent Profile:

- Edit: name, phone, bio, profile photo, languages spoken, WhatsApp number
- Change password with current password verification
- Notification preferences: Email / SMS / Portal alert per event type

PAGE 5 — COMPLETE SYSTEM WORKFLOW

5.1 End-to-End Customer Journey

Step 1 — Discovery:

A customer visits the public website through organic search, social media, a paid ad, a referral link, or direct URL. The site is fully SEO-optimized to maximize discoverability.

Step 2 — Browse & Search:

The customer uses search filters to find properties matching their requirements. They view property detail pages exploring photos, description, location map, amenities, and payment plans.

Step 3 — Lead Submission:

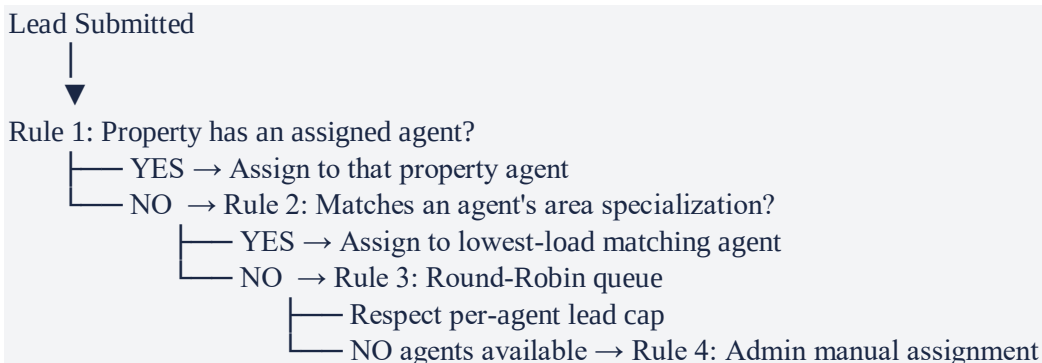
- Property Detail Page inquiry form (most common — pre-filled with property reference)
- Agent Profile Page contact form (lead tagged to that specific agent)
- General Contact Page form (type: General Inquiry)
- WhatsApp click-to-chat button (site-wide floating button)
- Phone call manually entered by Admin or Agent into the portal

Step 4 — Lead Record Created:

On form submission, a lead record is instantly created in the database with status New. The customer receives an automated confirmation email. The lead appears in the Admin Portal activity feed and All Leads table.

5.2 Lead Assignment Workflow

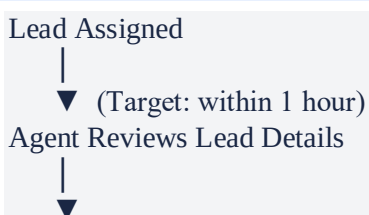
The Auto-Assignment Engine evaluates the following rules in sequence until a match is found:



Post-Assignment Triggers:

- Agent receives instant notification: Email + Portal alert + SMS (all configurable per agent)
- Lead status updates to Assigned; assignment timestamp logged in audit trail
- Admin sees the assignment in the activity feed in real time

5.3 Agent Contact & Lead Progression Workflow





5.4 Admin Oversight Workflow

At every stage, the Admin has real-time visibility without any action required by the agent to 'report.' All status changes, notes, and interactions are automatically surfaced in the Admin Portal.

Admin Intervention Points:

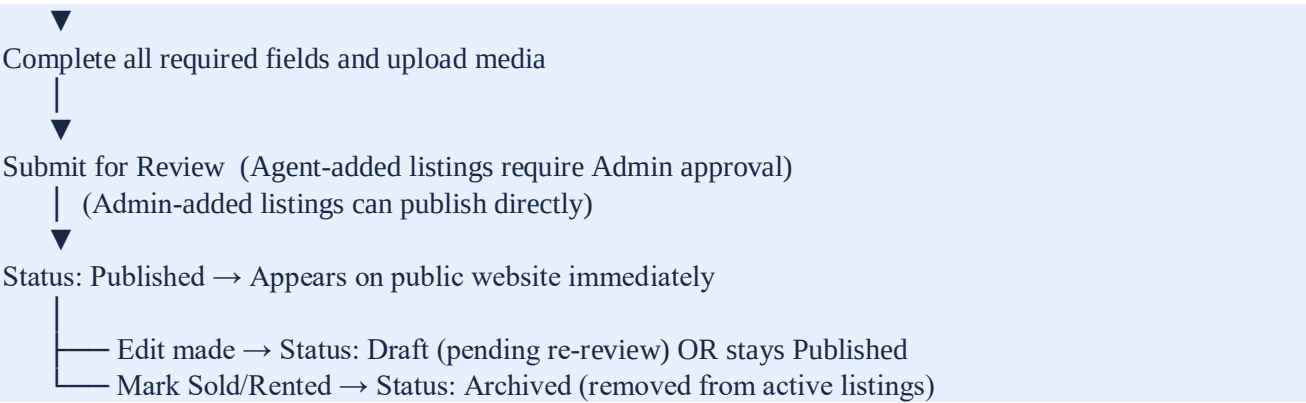
- Reassign a lead to a different agent at any time with one click
- Override lead status manually (e.g., recover a mistakenly closed lead)
- Add admin notes to any lead (visible to admin; not visible to agent or customer)
- Escalate a lead to a senior agent or branch manager with notification
- Archive or close stale leads in bulk using bulk-action tool
- Trigger escalation alert if no agent contact within configurable time window (default: 1 hour)

5.5 Notification Matrix

Event	Customer	Agent	Admin
Lead submitted by customer	✓ Confirmation email	—	Portal alert
Lead assigned to agent	—	✓ Email + SMS + Portal	Activity feed
Agent contacts customer	—	✓ Auto-logged	Status updated
Viewing scheduled	✓ Email + SMS reminder	✓ Calendar entry	Status updated
Lead status → Won	✓ Congratulations email	✓ Portal update	Report + alert
Lead status → Lost	—	✓ Portal update	Dashboard stat
No agent contact within 1 hour	—	✓ Reminder notification	Escalation alert
New property published	—	—	Activity feed
New customer registered	✓ Welcome email	—	Dashboard counter

5.6 Property Publication Workflow

Admin / Agent Adds Property (Status: Draft)



All content changes are logged in the Audit Trail with the admin's name, timestamp, and a before/after snapshot of every changed field. CMS edits are reflected on the public website immediately upon save (or on a scheduled publish date if configured).

5.7 Data Security & Access Control

- All portal access requires authenticated login with secure JWT session tokens
- Passwords stored as bcrypt hashes — never in plain text
- Two-Factor Authentication (2FA) available for Admin and Super Admin accounts
- All API endpoints protected with role-permission middleware (RBAC enforced server-side)
- HTTPS enforced site-wide with TLS/SSL certificates (auto-renewing via Let's Encrypt)
- Agent data is strictly scoped: agents cannot access other agents' leads or customer data
- IP logging on all admin logins for security audit trail
- GDPR-compliant data handling: customer data export (JSON/CSV) and deletion on request
- Regular automated database backups — daily, retained for 30 days, stored offsite

5.8 Scalability & Integration Roadmap

Phase	Scope
Phase 1 (MVP Launch)	Public website, Admin portal (property + lead + user mgmt), Agent portal (Lead CRM), Basic reporting
Phase 2 (Growth)	Portal API sync (Bayut, Property Finder), WhatsApp Business API, Google Calendar sync, Custom report builder, Agent mobile app
Phase 3 (Scale)	AI lead scoring & routing, Automated follow-up sequences, Commission management module, Multi-branch / multi-city support

Document prepared for: Real Estate Platform Project | Version 1.0 | May 2026

All modules, features, and workflows described herein are subject to refinement during the development planning phase.